Breast Cancer Tumor Classification:
A Comparative Study of Machine Learning Models

CPCS 433 - Artificial Intelligence Topics - Assignment #1

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1. Introduction

In this assignment I build three models to predict if a breast tumor is malignant (M) or benign (B). The models are Logistic Regression, Perceptron and Support Vector Machine (SVM). I wrote the code in a Python notebook (Google Colab/Jupyter) using pandas, matplotlib, seaborn and scikit-learn. This report shows the result of every step so the process can be understood without reading the code.

2. Dataset and loading

The dataset is *breast-cancer.csv*. I loaded it with pandas.read_csv. It has *569 rows* (patients) and *32 columns*: an *id* column, the *diagnosis* column (M or B), and 30 numeric features.

The 30 features come from 10 measurements of the cell nuclei (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension). Each measurement has three versions: the mean, the standard error (se) and the worst (largest) value. That gives 10 x 3 = 30 features.

The classes are a bit unbalanced: *357 benign (62.7%)* and *212 malignant (37.3%)*.

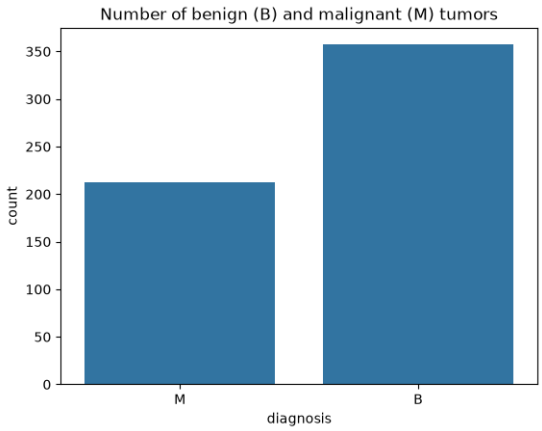


Figure 1: Number of benign (B) and malignant (M) tumors in the dataset.

3. Preprocessing

3.1 Missing values

I dropped the *id* column because it is only a patient number and does not help to predict the diagnosis. Then I counted the missing values in every column. The result was *0 missing values,* so no rows needed to be removed or filled.

3.2 Outliers

I checked outliers with the IQR rule. A value is an outlier if it is lower than Q1 - 1.5 x IQR or higher than Q3 + 1.5 x IQR. I found *608 outlier values out of 17,070* values (about 3.6%). The features with the most outliers were area_se (65), radius_se (38), perimeter_se (38) and area_worst (35). In total, 171 of the 569 rows have at least one outlier, and 114 of these 171 rows are malignant.

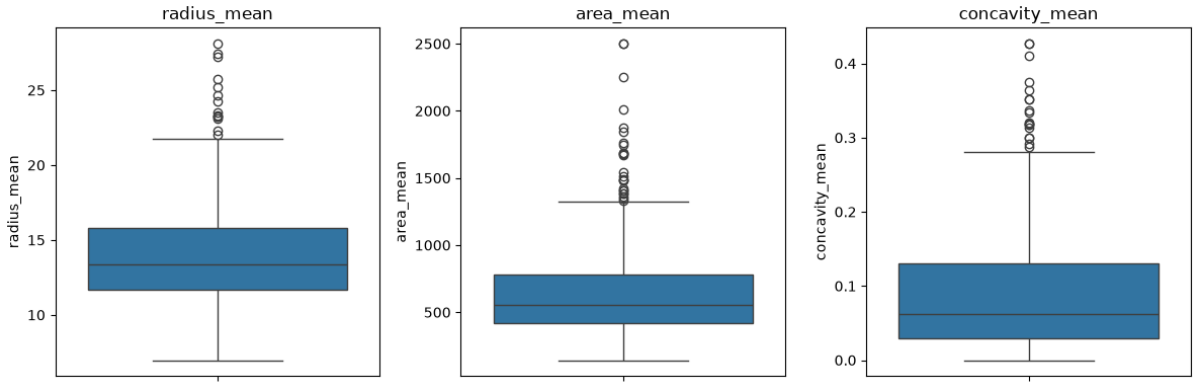


Figure 2: Boxplots of radius_mean, area_mean and concavity_mean. The dots above the boxes are outliers.

Note: I kept the outliers. Most of them belong to malignant tumors, so they look like real big or irregular tumors and not mistakes. Removing them would delete many useful malignant cases from a small dataset. I reduce their effect by scaling the features (section 3.4).

3.3 Encoding the diagnosis

The diagnosis is text (M or B), but the models need numbers. I changed *M to 1* and *B to 0*. After this, there are 357 zeros and 212 ones.

3.4 Scaling

The features have very different sizes (for example area is around 500-2500 but smoothness is around 0.1). I used *StandardScaler,* which makes every feature have mean 0 and standard deviation 1. I checked the scaled training data and the mean was 0 and the standard deviation was 1. To avoid data leakage, I split the data first and then fitted the scaler only on the training data (see next section).

4. Train/test split

I split the data *80% for training and 20% for testing* with ‘random_state=42’. I also used ‘stratify=y’ so the benign/malignant ratio is the same in both sets.

Set	Rows	Features
Training set	455	30
Test set	114	30

The scaler was fitted with the training data only, and then used to transform both the training and the test data. This way the test data gives no information (like its mean) to the training, so the test results are fair.

5. The models

5.1 Logistic Regression and Perceptron

I trained Logistic Regression (max_iter=1000) and Perceptron with ‘random_state=42’ on the scaled training data. Training accuracy was *0.9868* for Logistic Regression and *0.9780* for the Perceptron.

5.2 SVM with GridSearchCV

For the SVM, I used GridSearchCV with 5-fold cross validation on the training set. I tried *C = 0.1, 1, 10, 100* and *kernel = linear, rbf, poly, sigmoid*. That is 16 combinations.

Best parameters: C = 1 and kernel = 'rbf', with a cross-validation accuracy of *0.9758.*

Rank	C	Kernel	Mean CV accuracy
1	1	rbf	0.9758
2	10	rbf	0.9736
3 (tie)	10	linear	0.9692
3 (tie)	0.1	linear	0.9692
5 (tie)	1	linear	0.9626
5 (tie)	100	poly	0.9626

Table 1: Best 6 results of the grid search (the full table is in the notebook).

6. Evaluation on the test set

I used the test set (114 patients) to evaluate the three models. Below are the classification reports (precision, recall and F1-score for each class). The SVM uses the best parameters.

Logistic Regression

Class	Precision	Recall	F1-score	Support
Benign (0)	0.96	0.99	0.97	72
Malignant (1)	0.97	0.93	0.95	42
Accuracy			0.96	114

Perceptron

Class	Precision	Recall	F1-score	Support
Benign (0)	0.97	0.99	0.98	72
Malignant (1)	0.98	0.95	0.96	42
Accuracy			0.97	114

SVM (C = 1, rbf)

Class	Precision	Recall	F1-score	Support
Benign (0)	0.96	1.00	0.98	72
Malignant (1)	1.00	0.93	0.96	42
Accuracy			0.97	114

Table 2: Classification reports on the test set.

Model	Test accuracy
Logistic Regression	0.9649
Perceptron	0.9737
SVM	0.9737

Table 3: Exact test accuracy of each model.

All three models have a high accuracy (96.5% to 97.4%). The Perceptron and the SVM are the best with 0.9737.

7. Confusion matrices

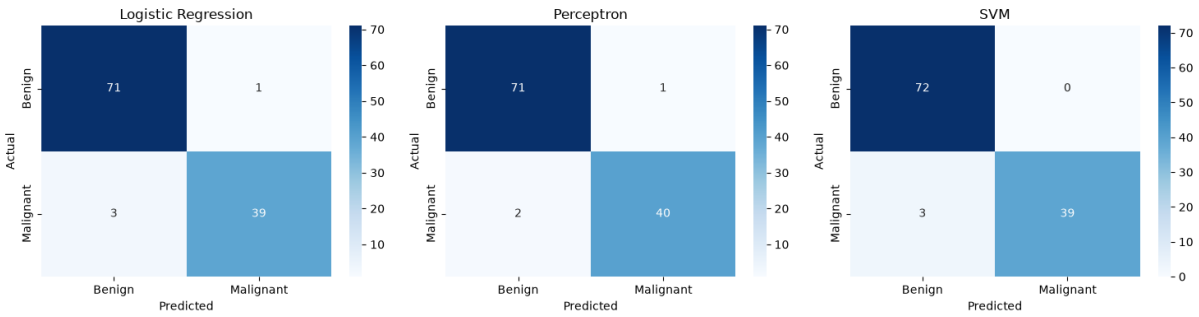


Figure 3: Confusion matrices of the three models on the test set.

Model	**Correct benign**	**Benign predicted as malignant**	**Malignant predicted as benign**	**Correct malignant**
Logistic Regression	71	1	3	39
Perceptron	71	1	2	40
SVM	72	0	3	39

Table 4: Numbers from the confusion matrices.

The Logistic Regression made 4 mistakes, and the Perceptron and the SVM made 3 mistakes each. The most serious mistake is a malignant tumor predicted as benign (a missed cancer). The Perceptron missed only 2, and the other two models missed 3. The SVM had no false alarms (0 benign predicted as malignant).

8. ROC curves

The ROC curve shows the true positive rate against the false positive rate for different thresholds. A curve closer to the top-left corner is better, and the AUC (area under the curve) is closer to 1 for a better model. For the Logistic Regression I used the predicted probabilities, and for the Perceptron and the SVM I used decision_function.

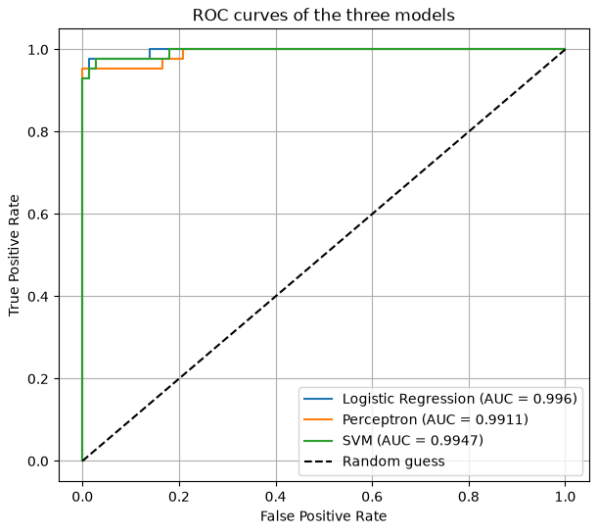


Figure 4: ROC curves of the three models on one graph.

Model	**AUC**
Logistic Regression	0.9960
Perceptron	0.9911
SVM	0.9947

Table 5: AUC of each model.

All three AUC values are above 0.99, so all models separate the two classes very well. Logistic Regression has the highest AUC (0.9960), then the SVM (0.9947) and the Perceptron (0.9911).

9. Decision boundaries with PCA

The data has 30 features, so it is not possible to draw a decision boundary in 30 dimensions. For this plot only, I reduced the scaled data to *2 dimensions with PCA*. The two components keep 44.6% + 18.5% = *63.1%* of the variance, so some information is lost.

The earlier models were trained with 30 features, so they cannot work on 2D points. Because of this, I retrained the Logistic Regression, Perceptron and SVM on the 2D PCA data. The SVM used the best parameters from the grid search (C = 1, rbf). In the plots, benign tumors are blue, malignant tumors are red, and the black line is the decision boundary.

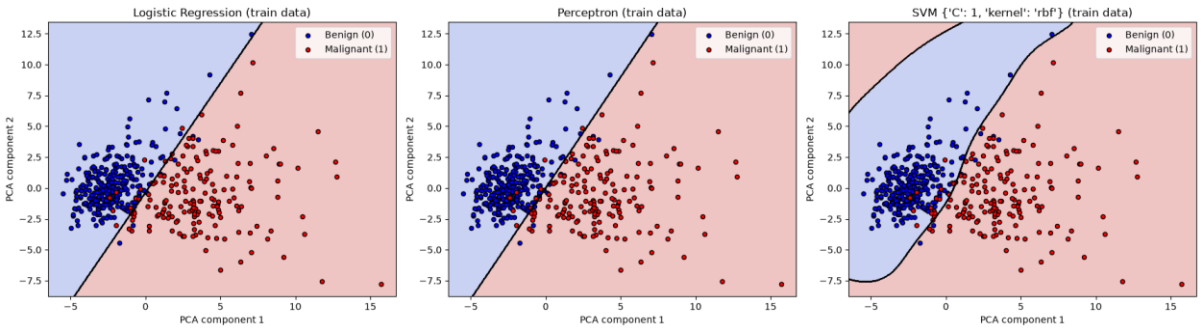


Figure 5: Decision boundaries on the training data (2D PCA).

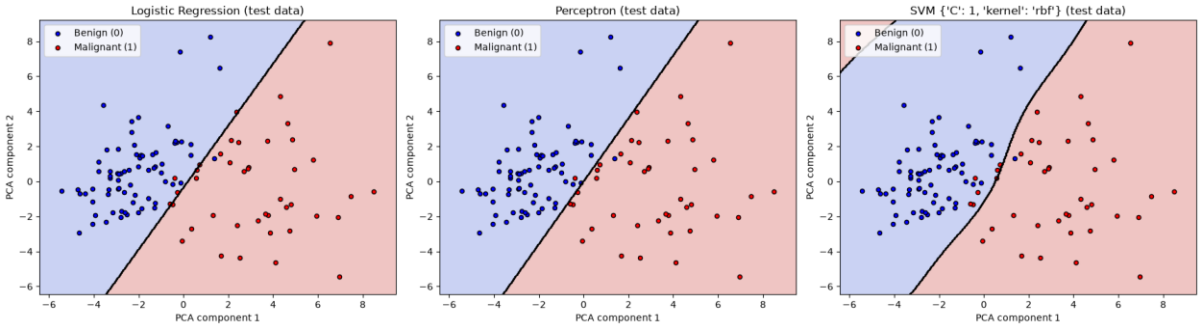


Figure 6: Decision boundaries on the test data (2D PCA).

Model	Test accuracy (30 features)	Test accuracy (2D PCA)
Logistic Regression	0.9649	0.9474
Perceptron	0.9737	0.9825
SVM	0.9737	0.9298

Table 6: Test accuracy of the models with all features and with only 2 PCA components.

Logistic Regression and the Perceptron have a straight line boundary, and the SVM with the rbf kernel has a curved boundary. The two classes overlap a little in the middle, which is where the mistakes happen. The 2D accuracies are different from the 30-feature ones because PCA loses some information, so these plots are only for visualization.

10. Conclusion

The best SVM parameters were *C = 1* and *kernel = rbf*. This is the final comparison on the test set:

Model	Accuracy	AUC	Malignant recall	Malignant precision	Mistakes
Logistic Regression	0.9649	0.9960	0.93	0.97	4
Perceptron	0.9737	0.9911	0.95	0.98	3
SVM (C=1, rbf)	0.9737	0.9947	0.93	1.00	3

Table 7: Final comparison of the three models.

- All three models work very well on this dataset: test accuracy is between 96.5% and 97.4% and the AUC is above 0.99.
- The Perceptron and the SVM have the best accuracy (0.9737). The Perceptron found the most malignant tumors (recall 0.95), and the SVM never gave a false alarm (malignant precision 1.00).
- Logistic Regression has the best AUC (0.9960) but the lowest accuracy (0.9649).
- Missing a malignant tumor is worse than a false alarm, so the Perceptron looks best on this point (2 missed against 3 for the others).
- The test set is small (114 patients), so the differences of 1 or 2 patients may be only luck from the split.

Overall: a simple linear model is already enough for this data. I would choose the SVM because it has a very high AUC and the best precision, but the Perceptron and Logistic Regression are simpler and almost as good.